



Unit I: Descriptive Statistics

Measures of Central Tendency, Dispersion,
Skewness & Kurtosis

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1 Introduction to Descriptive Statistics

Descriptive statistics provide methods to summarize, organize, and present data in a meaningful way. For an engineer, they are the first step in understanding measurement variability, product quality, and process behaviour. The three key aspects we quantify are:

- **Central tendency** – a typical or representative value.
- **Dispersion** – the spread or variability.
- **Shape** – symmetry (skewness) and tail heaviness (kurtosis).

1.1 Types of Data

Data can be classified as:

- **Qualitative (categorical)**: non-numeric labels (e.g., machine status: on/off, type of defect).
- **Quantitative (numerical)**: numeric measurements.
 - *Discrete*: countable (e.g., number of defective items, number of cylinders).
 - *Continuous*: measurable on a continuum (e.g., voltage, temperature, diameter).

1.2 Listing of Data

Data can be presented in two fundamental ways:

1. **Raw (ungrouped) data**: a simple list of individual observations.
2. **Frequency distribution**: a table that shows distinct values (or class intervals) and how often they occur. This summarises large datasets.

In this unit, we work with both forms, using appropriate formulas for each.

2 Measures of Central Tendency

A measure of central tendency is a single value that attempts to describe the centre of a data set.

2.1 Arithmetic Mean

2.1.1 For Individual Observations

For n values $x_1, x_2, \dots, x_n$,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (1)$$

2.1.2 For Discrete Frequency Distributions

If values x_i occur with frequencies f_i ($i = 1, \dots, k$) and $N = \sum f_i$,

$$\bar{x} = \frac{\sum_{i=1}^k f_i x_i}{N}. \quad (2)$$

2.1.3 For Continuous Grouped Data

When data are grouped into class intervals, let x_i be the midpoint of the i -th class and f_i its frequency. The mean is calculated using the same formula as above.

Shortcut (Assumed Mean) Method Choose an assumed mean A (often a central midpoint). Define $d_i = x_i - A$. Then

$$\bar{x} = A + \frac{\sum f_i d_i}{N}. \quad (3)$$

Step-Deviation Method If all class widths are equal to h , set $u_i = \frac{x_i - A}{h}$. Then

$$\bar{x} = A + h \cdot \frac{\sum f_i u_i}{N}. \quad (4)$$

Weighted Arithmetic Mean When observations carry different weights w_i ,

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}. \quad (5)$$

Solved Example

The breaking strength (in MPa) of 12 identical steel rods tested in a lab is:

485, 492, 500, 495, 488, 505, 498, 490, 502, 489, 500, 491.

Calculate the mean tensile strength. If the design requires a mean strength of at least 495 MPa, does this batch meet the requirement?

Solution:

$$\begin{aligned} \bar{x} &= \frac{485 + 492 + 500 + 495 + 488 + 505 + 498 + 490 + 502 + 489 + 500 + 491}{12} \\ &= \frac{5935}{12} \approx 494.58 \text{ MPa.} \end{aligned}$$

The mean is slightly below 495 MPa, so the batch does **not** meet the requirement.

Solved Example

The diameters (in mm) of 100 ball bearings are summarised below.

Diameter (mm)	Frequency
10 – 12	6
12 – 14	18
14 – 16	34
16 – 18	28
18 – 20	14

Using the step-deviation method with assumed mean $A = 15$ (midpoint of 14–16), compute the mean diameter.

Solution: $h = 2$. Midpoints: 11, 13, 15, 17, 19.

$$u_i = \frac{x_i - 15}{2} : \quad -2, -1, 0, 1, 2.$$

x_i	f_i	u_i	$f_i u_i$
11	6	-2	-12
13	18	-1	-18
15	34	0	0
17	28	1	28
19	14	2	28
$N = 100$			$\sum f_i u_i = 26$

$$\bar{x} = A + h \frac{\sum f_i u_i}{N} = 15 + 2 \cdot \frac{26}{100} = 15 + 0.52 = 15.52 \text{ mm.}$$

2.2 Median

The median is the middle value of ordered data.

2.2.1 Ungrouped Data

For n odd, median $= x_{(\frac{n+1}{2})}$; for n even, median $= \frac{1}{2}(x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)})$.

2.2.2 Grouped Continuous Data

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f_m} \right) \times w,$$

where L = lower true boundary of median class, F = cumulative frequency before median class, f_m = frequency of median class, w = class width.

Solved Example

For the ball bearing data in Example 2.1.3, find the median diameter.

Solution: Cumulative frequencies:

Class	f	Cum.
10 – 12	6	6
12 – 14	18	24
14 – 16	34	58
16 – 18	28	86
18 – 20	14	100

$N/2 = 50$. The median class is 14–16 (cum. freq. 58 exceeds 50). $L = 14$, $F = 24$,

$$f_m = 34, w = 2.$$

$$\text{Median} = 14 + \left(\frac{50 - 24}{34} \right) \times 2 = 14 + \frac{26}{34} \times 2 \approx 14 + 1.529 = 15.529 \text{ mm.}$$

2.3 Mode

The mode is the most frequent value.

2.3.1 Grouped Continuous Data

$$\text{Mode} = L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) \times w,$$

where L = lower boundary of modal class, f_m = frequency of modal class, f_1 = frequency of class before, f_2 = frequency of class after.

Solved Example

Find the modal diameter for the ball bearing data.

Solution: The highest frequency is 34 (class 14–16). Thus $L = 14$, $f_m = 34$, $f_1 = 18$, $f_2 = 28$, $w = 2$.

$$\text{Mode} = 14 + \left(\frac{34 - 18}{2 \cdot 34 - 18 - 28} \right) \times 2 = 14 + \left(\frac{16}{68 - 46} \right) \times 2 = 14 + \frac{16}{11} \approx 15.455 \text{ mm.}$$

2.4 Empirical Relationship (Mean, Median, Mode)

For a moderately skewed unimodal distribution:

$$\text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median}).$$

3 Measures of Dispersion

Dispersion indicates the spread or variability. An engineer must know how consistent a process is.

3.1 Range

$$R = X_{\max} - X_{\min}.$$

$$\text{Coefficient of Range} = \frac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}.$$

3.2 Mean Deviation (Average Absolute Deviation)

$$MD_{\text{mean}} = \frac{\sum f_i |x_i - \bar{x}|}{N}, \quad MD_{\text{median}} = \frac{\sum f_i |x_i - \text{Med}|}{N}.$$

3.3 Standard Deviation and Variance

Variance is the average squared deviation from the mean.

3.3.1 Population Standard Deviation

Individual observations:

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}. \quad (6)$$

Frequency distribution:

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}. \quad (7)$$

3.3.2 Shortcut (Assumed Mean) Formula

Let $d_i = x_i - A$. Then

$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N} \right)^2}. \quad (8)$$

3.3.3 Step-Deviation Formula

For equal class width h , $u_i = \frac{x_i - A}{h}$,

$$\sigma = h \cdot \sqrt{\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N} \right)^2}. \quad (9)$$

3.3.4 Coefficient of Variation (CV)

$$CV = \frac{\sigma}{\bar{x}} \times 100\%.$$

Solved Example

A manufacturer records the lifetimes (hours) of 10 LED bulbs: 1010, 980, 1005, 1020, 990, 1015, 985, 1000, 995, 1010. Compute the standard deviation and coefficient of variation. Interpret the consistency.

Solution: Mean $\bar{x} = 1001$ hrs. Deviations: 9, -21, 4, 19, -11, 14, -16, -1, -6, 9.

$$\sum (x_i - \bar{x})^2 = 81 + 441 + 16 + 361 + 121 + 196 + 256 + 1 + 36 + 81 = 1590.$$

$\sigma = \sqrt{1590/10} \approx 12.61$ hrs. $CV = (12.61/1001) \times 100\% \approx 1.26\%$. The low CV indicates very consistent lifetimes.

Solved Example

Use the step-deviation method to compute the standard deviation of the ball bearing diameters from Example 2.1.3.

Solution: We already have $A = 15$, $h = 2$, $u_i = -2, -1, 0, 1, 2$, $\sum f_i u_i = 26$. Now compute $f_i u_i^2$:

x_i	f_i	u_i	$f_i u_i^2$
11	6	-2	24
13	18	-1	18
15	34	0	0
17	28	1	28
19	14	2	56
	100		$\sum f_i u_i^2 = 126$

$$\sigma = h \sqrt{\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N} \right)^2} = 2 \sqrt{\frac{126}{100} - \left(\frac{26}{100} \right)^2} = 2 \sqrt{1.1924} \approx 2.18 \text{ mm.}$$

4 Moments: Skewness and Kurtosis

All moments shown are central moments, i.e., moments about the arithmetic mean. We illustrate how they capture skewness (asymmetry) and kurtosis (peakedness/tails).

4.1 Central Moments (Definition)

For a frequency distribution with values x_i , frequencies f_i and $N = \sum f_i$, the r -th central moment is

$$m_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^r, \quad \bar{x} = \frac{\sum f_i x_i}{N}.$$

Key properties:

$$m_1 = 0, \quad m_2 = \sigma^2.$$

Using the moments, we define two key metrics γ_1 & β_2

$$\gamma_1 = \frac{m_3}{\sigma^3}, \quad \& \quad \beta_2 = \frac{m_4}{\sigma^4}.$$

Here, γ_1 gives information about the skewness of the data and β_2 tells us about the kurtosis of the data.

$$\gamma_1 \text{ is } \begin{cases} > 0 & : \text{Positively Skewed} \\ = 0 & : \text{Symmetric,} \\ < 0 & : \text{Negatively Skewed.} \end{cases}$$

$$\beta_2 \text{ is } \begin{cases} > 3 & : \text{Leptokurtic} \\ = 3 & : \text{Mesokurtic,} \\ < 3 & : \text{Platykurtic.} \end{cases}$$

4.2 Example 1 – Right-skewed frequency data

Solved Example

The number of defective components per box is recorded for 100 boxes.

Defects (x_i)	0	1	2	3	4
Boxes (f_i)	20	35	25	15	5

Compute the first four central moments, γ_1 and β_2 . Plot the histogram.

Solution:

$$\bar{x} = \frac{0 \cdot 20 + 1 \cdot 35 + 2 \cdot 25 + 3 \cdot 15 + 4 \cdot 5}{100} = \frac{0 + 35 + 50 + 45 + 20}{100} = \frac{150}{100} = 1.5.$$

Table 1: Central moments for the defect data

x_i	f_i	$d_i = x_i - 1.5$	$f_i d_i^2$	$f_i d_i^3$	$f_i d_i^4$
0	20	-1.5	$20 \times 2.25 = 45.0$	$20 \times (-3.375) = -67.5$	$20 \times 5.0625 = 101.25$
1	35	-0.5	$35 \times 0.25 = 8.75$	$35 \times (-0.125) = -4.375$	$35 \times 0.0625 = 2.1875$
2	25	0.5	$25 \times 0.25 = 6.25$	$25 \times 0.125 = 3.125$	$25 \times 0.0625 = 1.5625$
3	15	1.5	$15 \times 2.25 = 33.75$	$15 \times 3.375 = 50.625$	$15 \times 5.0625 = 75.9375$
4	5	2.5	$5 \times 6.25 = 31.25$	$5 \times 15.625 = 78.125$	$5 \times 39.0625 = 195.3125$
	100		$\sum f_i d_i^2 = 125.0$	$\sum f_i d_i^3 = 60.0$	$\sum f_i d_i^4 = 376.25$

$$m_2 = \frac{125}{100} = 1.25, \quad m_3 = \frac{60}{100} = 0.6, \quad m_4 = \frac{376.25}{100} = 3.7625.$$

$$\gamma_1 = \frac{m_3}{m_2^{3/2}} = \frac{0.6}{1.25^{1.5}} = \frac{0.6}{1.3975} \approx 0.429 \quad (\text{positively skewed}).$$

$$\beta_2 = \frac{m_4}{m_2^2} = \frac{3.7625}{1.5625} \approx 2.408, \quad \gamma_2 = 2.408 - 3 = -0.592 \quad (\text{platykurtic}).$$

The histogram (Figure 1) confirms the right-skew: the tail extends further to the right.

4.3 Example 2 – Symmetric frequency data

Solved Example

A fair coin is tossed 5 times; the number of heads obtained in 320 repetitions is:

Heads (x_i)	0	1	2	3	4	5
Frequency (f_i)	10	50	100	100	50	10

Compute the central moments, γ_1 and β_2 . Interpret.

Solution:

$$\bar{x} = \frac{0 \cdot 10 + 1 \cdot 50 + 2 \cdot 100 + 3 \cdot 100 + 4 \cdot 50 + 5 \cdot 10}{320} = \frac{800}{320} = 2.5.$$

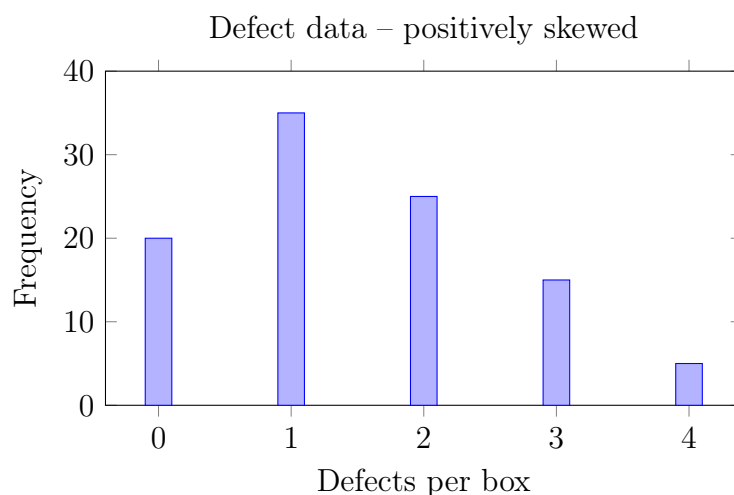


Figure 1: Histogram showing positive skew.

Table 2: Central moments for coin-toss data

x_i	f_i	$d_i = x_i - 2.5$	$f_i d_i^2$	$f_i d_i^3$	$f_i d_i^4$
0	10	-2.5	$10 \times 6.25 = 62.5$	$10 \times (-15.625) = -156.25$	$10 \times 39.0625 = 390.625$
1	50	-1.5	$50 \times 2.25 = 112.5$	$50 \times (-3.375) = -168.75$	$50 \times 5.0625 = 253.125$
2	100	-0.5	$100 \times 0.25 = 25.0$	$100 \times (-0.125) = -12.5$	$100 \times 0.0625 = 6.25$
3	100	0.5	$100 \times 0.25 = 25.0$	$100 \times 0.125 = 12.5$	$100 \times 0.0625 = 6.25$
4	50	1.5	$50 \times 2.25 = 112.5$	$50 \times 3.375 = 168.75$	$50 \times 5.0625 = 253.125$
5	10	2.5	$10 \times 6.25 = 62.5$	$10 \times 15.625 = 156.25$	$10 \times 39.0625 = 390.625$
320			$\Sigma = 400.0$	$\Sigma = 0$	$\Sigma = 1300.0$

$$m_2 = \frac{400}{320} = 1.25, \quad m_3 = 0, \quad m_4 = \frac{1300}{320} = 4.0625.$$

$$\gamma_1 = 0 \quad (\text{perfect symmetry}), \quad \beta_2 = \frac{4.0625}{1.5625} \approx 2.6, \quad \gamma_2 = -0.4 \quad (\text{platykurtic}).$$

The histogram (Figure 2) is symmetric and somewhat flat.

4.4 Graphical Understanding of Skewness

Skewness is shown by the direction of the longer tail. If the longer tail is on the right side of the graph, we say it is positively skewed, and if the left tail is longer, the data are called negatively skewed.

4.5 Graphical Understanding of Kurtosis

Kurtosis compares the peak and the weight of the tail with the normal distribution ($\beta_2 = 3$). If the data have higher peak and longer tail compared to bell shaped curved then it is called Leptokurtic, and if the data have flatter peak and longer tail compared to bell shaped curve then it is called Platykurtic. A bell shaped normal curve is called Mesokurtic.

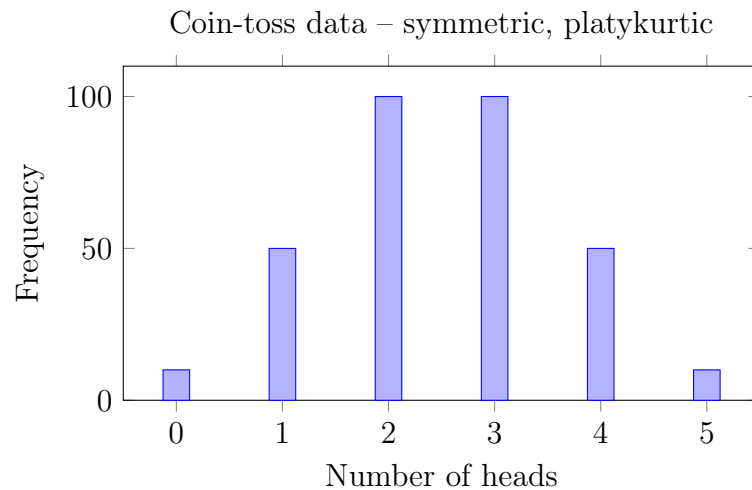


Figure 2: Symmetric distribution.

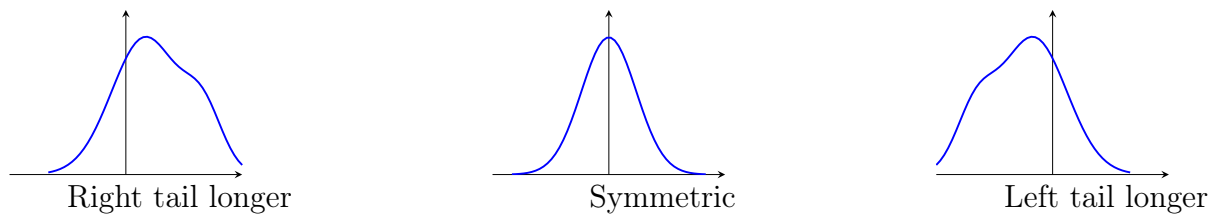


Figure 3: Left: positive skew. Middle: symmetric. Right: negative skew.

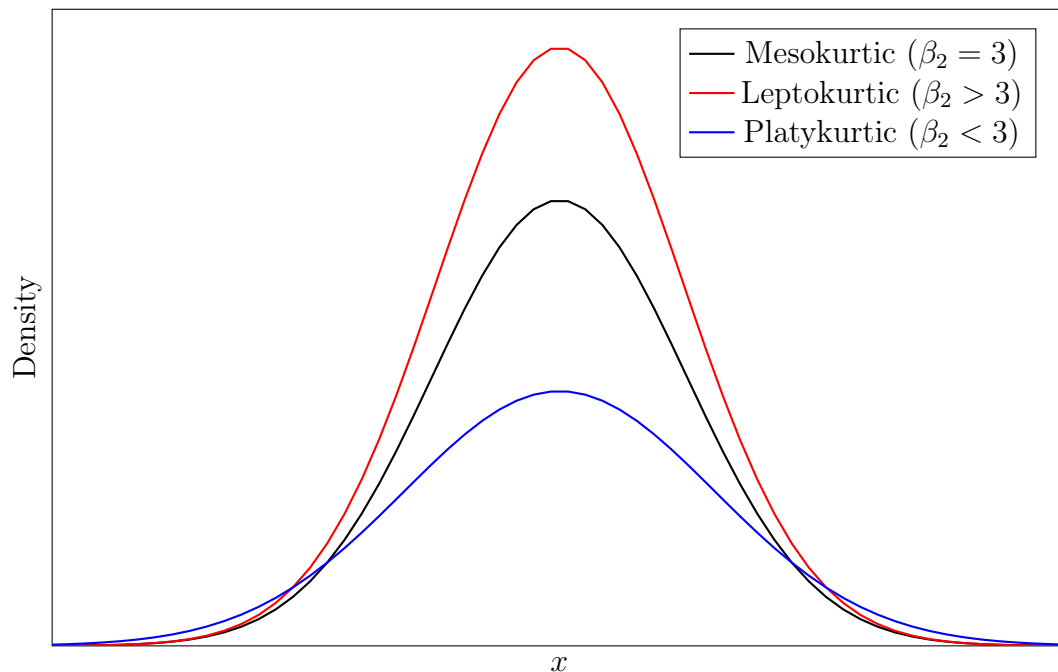


Figure 4: Kurtosis types compared. Leptokurtic has a sharper peak and heavier tails; platykurtic is flatter with thinner tails.

4.6 Example 3 – Symmetric grouped continuous data

Solved Example

The diameters (in mm) of 100 shafts are distributed symmetrically as:

Diameter (mm)	Frequency (f_i)
10 – 12	10
12 – 14	20
14 – 16	40
16 – 18	20
18 – 20	10

Compute the central moments, γ_1 and β_2 using the step-deviation method. Because the data are symmetric, the mean equals the midpoint of the central class, i.e., $\bar{x} = 15$.

Solution: Midpoints x_i : 11, 13, 15, 17, 19. Let $A = 15$, $h = 2$. Then $u_i = \frac{x_i - 15}{2} = -2, -1, 0, 1, 2$.

x_i	f_i	u_i	$f_i u_i$	$f_i u_i^2$	$f_i u_i^4$
11	10	-2	-20	40	160
13	20	-1	-20	20	20
15	40	0	0	0	0
17	20	1	20	20	20
19	10	2	20	40	160
	100		$\sum f_i u_i = 0$	$\sum f_i u_i^2 = 120$	$\sum f_i u_i^4 = 360$

Since $\sum f_i u_i = 0$, the raw u -moments equal the central u -moments:

$$m_u(2) = \frac{120}{100} = 1.2, \quad m_u(4) = \frac{360}{100} = 3.6.$$

Convert to original units:

$$m_2 = h^2 m_u(2) = 4 \times 1.2 = 4.8, \quad m_4 = h^4 m_u(4) = 16 \times 3.6 = 57.6.$$

$$\gamma_1 = 0 \quad (\text{symmetric}), \quad \beta_2 = \frac{m_4}{m_2^2} = \frac{57.6}{4.8^2} = \frac{57.6}{23.04} = 2.5, \quad \gamma_2 = -0.5 \quad (\text{platykurtic}).$$

The histogram (Figure 5) shows a symmetric, moderately flat shape.

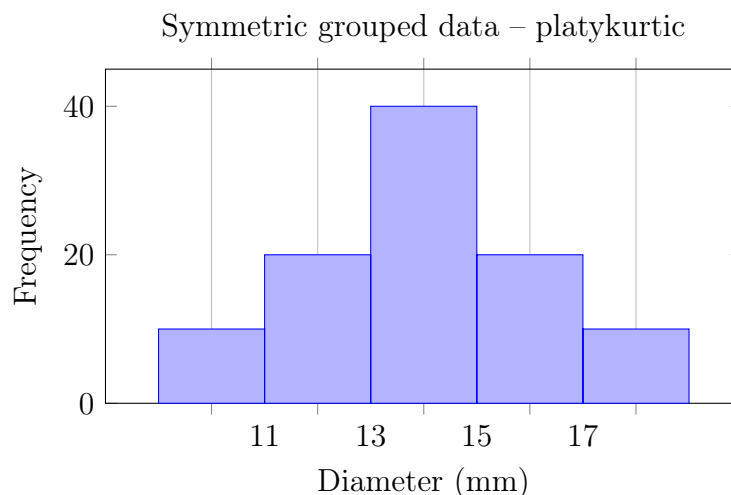


Figure 5: Symmetric, platykurtic distribution.

5 Summary Table of Descriptive Measures

Table 3: Summary of descriptive statistics

Aspect	Measure	Key Formula
Central Tendency	Arithmetic Mean	$\bar{x} = \frac{\sum f_i x_i}{N}$
	Median (grouped)	$L + \left(\frac{N/2 - F}{f_m} \right) w$
	Mode (grouped)	$L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) w$
Dispersion	Range	$X_{\max} - X_{\min}$
	Mean Deviation	$\frac{\sum f_i x_i - \bar{x} }{N}$
	Standard Deviation	$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}$
	Coefficient of Variation	$\frac{\sigma}{\bar{x}} \times 100\%$
Skewness	Pearson's	$\frac{3(\text{Mean} - \text{Median})}{\sigma}$
	Moment γ_1	m_3 / σ^3
Kurtosis	β_2	m_4 / m_2^2
	Excess γ_2	$\beta_2 - 3$

6 Practice Problems

- The following data are the compression strengths (in MPa) of 15 concrete cubes: 28.5, 31.2, 29.8, 33.1, 30.0, 32.4, 29.5, 31.0, 30.8, 32.0, 28.9, 31.5, 30.3, 33.0, 29.0. Compute mean, median, mode, range, mean deviation about the mean, standard deviation, and coefficient of variation.
- The frequency distribution of the time to failure (in hours) of 80 electronic components is:

Time (hrs)	Components
0–50	5
50–100	12
100–150	28
150–200	22
200–250	10
250–300	3

Using the step-deviation method, compute the mean, median, mode, standard deviation, and Pearson's coefficient of skewness.

- For the repair time data in Example ??, compute the fourth central moment and the kurtosis β_2 . (Hint: you already have the mean and deviations.)
- A dataset of 500 network latency measurements yields $m_2 = 4.0 \text{ ms}^2$ and $m_4 = 64.0 \text{ ms}^4$. Find the kurtosis and discuss whether the latency distribution shows heavy tails.
- Explain why standard deviation is preferred over mean deviation in quality control engineering.